

Q2: 2023 Backcast Revenue Calculation

GCV Solar + Storage Valuation — NYISO N.Y.C. Zone

E3 Applicant Exercise

2026

Inputs and Notation

All revenue is calculated from hourly 2023 NYISO N.Y.C. zone real-time prices. For hour t , let p_t be the real-time energy price (\$/MWh) and s_t be the normalized solar production shape. Solar output is assumed to be available for the full hour, so MW and MWh are numerically equivalent for each hourly interval.

Parameter	Value
Solar size	300 MW-DC
Solar inverter limit	250 MW-AC
Storage power capacity	50 MW
Storage energy capacity	200 MWh
Storage duration	200 MWh/50 MW = 4 hours
Round-trip efficiency	100% per provided input
Dispatch simplification	One full charge/discharge cycle per day with perfect real-time foresight

Q2a. Standalone Solar Revenue

Hourly solar generation is calculated from the normalized production shape and capped at the inverter limit:

$$G_t = \min(s_t \times 300 \text{ MW}, 250 \text{ MW})$$

Annual backcast solar revenue is the sum of hourly delivered energy times the real-time price:

$$R_{\text{solar}} = \sum_{t=1}^{8760} G_t \times p_t$$

Using the 2023 hourly data:

$$\sum_t G_t = 551,454 \text{ MWh}$$

$$R_{\text{solar}} = \$18,910,425$$

$$\text{Solar capture price} = \frac{\$18,910,425}{551,454 \text{ MWh}} = \$34.29/\text{MWh}$$

The inverter cap clips 372 MWh, equal to 0.07% of gross solar generation, so clipping has a negligible effect on standalone solar revenue.

Q2b. Standalone Storage Revenue

Storage is modeled as daily energy arbitrage. For each day d , the battery charges during the four lowest-priced hours and discharges during the four highest-priced hours:

$$R_{\text{storage},d} = 50 \text{ MW} \times \left(\sum_{h \in H_d^{\text{high}}} p_h - \sum_{h \in H_d^{\text{low}}} p_h \right)$$

where H_d^{high} is the set of the day's four highest-price hours and H_d^{low} is the set of the day's four lowest-price hours.

Annual standalone storage revenue is:

$$R_{\text{storage}} = \sum_{d=1}^{365} R_{\text{storage},d} = \$2,473,753$$

Normalized by installed capacity:

$$\frac{\$2,473,753}{50,000 \text{ kW}} = \$49.48/\text{kW-year}$$

$$\frac{\$2,473,753}{200,000 \text{ kWh}} = \$12.37/\text{kWh-year}$$

Q2c. Combined Solar + Storage Revenue

The combined case keeps the same solar export calculation as standalone solar. Storage first charges from otherwise-clipped solar energy at zero opportunity cost, then charges any remaining daily energy requirement from the grid during low-price hours.

For each hour, clipped solar is:

$$C_t = \max(s_t \times 300 \text{ MW} - 250 \text{ MW}, 0)$$

Daily storage charging from clipped solar is limited by storage power and daily energy capacity:

$$E_d^{\text{clip}} = \min \left(200 \text{ MWh}, \sum_{t \in d} \min(C_t, 50 \text{ MW}) \right)$$

The remaining daily storage charge is bought from the grid:

$$E_d^{\text{grid}} = 200 \text{ MWh} - E_d^{\text{clip}}$$

Storage then discharges during the day's four highest-price hours, as in the standalone storage case. Across 2023, the model uses:

$$\sum_d E_d^{\text{clip}} = 372 \text{ MWh}$$

$$\sum_d E_d^{\text{grid}} = 72,628 \text{ MWh}$$

The resulting combined revenues are:

$$R_{\text{combined solar}} = \$18,910,425$$

$$R_{\text{combined storage}} = \$2,480,611$$

$$R_{\text{combined total}} = \$18,910,425 + \$2,480,611 = \$21,391,036$$

The interaction benefit versus building standalone solar and standalone storage separately is:

$$R_{\text{combined total}} - (R_{\text{solar}} + R_{\text{storage}}) = \$21,391,036 - (\$18,910,425 + \$2,473,753) = \$6,857$$

$$\frac{\$6,857}{\$21,384,178} = 0.032\%$$

Q2d. Reserve (Headroom/Footroom) Scenario — Extension Beyond the Base Exercise

As a further scenario, we estimate additional revenue each configuration could earn by offering NYISO operating reserve capacity, at an assumed reserve price of \$1.25/MW-hr (midpoint of the \$1–1.5/MW range). *Headroom* is capacity available to increase output on call (up-reserve); *footroom* is capacity available to decrease output on call (down-reserve).

Solar can only offer footroom — it is already running at its irradiance-limited maximum, so it has no spare capacity to ramp up. Footroom each hour equals that hour's delivered MW, since output could be curtailed to zero on instruction:

$$R_{\text{solar reserve}} = \$1.25 \times \sum_t G_t = \$1.25 \times 551,454 \text{ MWh} = \$689,318$$

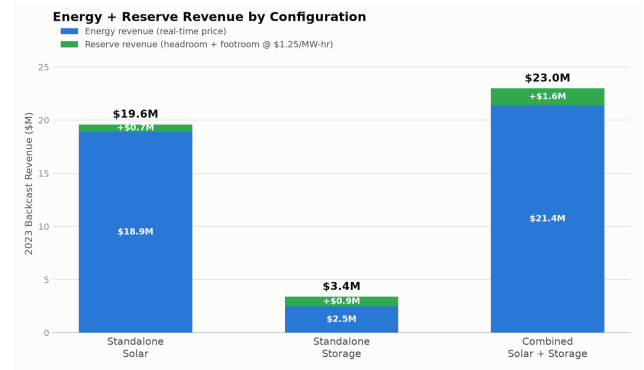
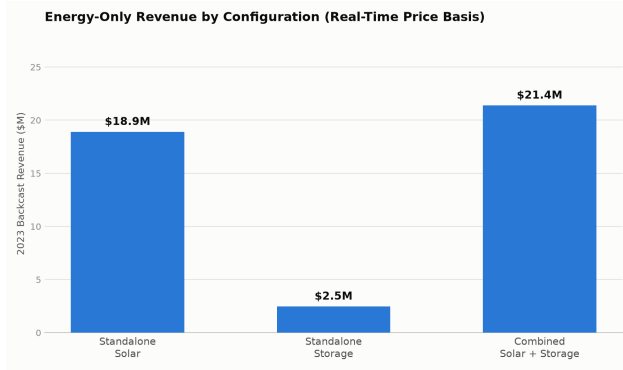
Storage reuses the Q2b/Q2c daily charge/discharge schedule (4 lowest-price hours charge, 4 highest-price hours discharge, 16 hours idle). While charging it can offer 50 MW of footroom (no headroom, since it is already using its full power to charge); while discharging the reverse (50 MW headroom, no footroom); while idle it can offer the full 50 MW of *both* headroom and footroom simultaneously:

$$R_{\text{storage reserve}} = \$1.25 \times (365,000 + 365,000) \text{ MW-hr} = \$912,500$$

The combined system sums solar’s footroom and storage’s headroom/footroom, since the combined case’s dispatch *timing* is unchanged from Q2b/Q2c — only the charging cost source (clipped solar vs. grid) differs:

$$R_{\text{combined reserve}} = \$689,318 + \$912,500 = \$1,601,818$$

Configuration	Energy-Only	Reserve	Energy + Reserve	Reserve Uplift
Standalone solar	\$18.910M	\$0.689M	\$19.600M	3.6%
Standalone storage	\$2.474M	\$0.913M	\$3.386M	36.9%
Combined solar + storage	\$21.391M	\$1.602M	\$22.993M	7.5%



Storage benefits by far the most in relative terms (+36.9%) because its reserve capacity comes free alongside energy dispatch and, while idle (16 of 24 hours/day), it can stack both headroom and footroom at the same time. Solar’s uplift is small in absolute dollars because footroom only applies to hours it is already delivering, but it is non-trivial (+3.6%) simply because solar runs so many hours. This scenario is illustrative: real NYISO reserve products have stricter response-time, minimum-run, and co-optimization rules than modeled here, and a resource cannot literally be paid full price for headroom and footroom at the same instant if doing so double-counts the same physical capacity across products — a more rigorous treatment is one of the natural extensions noted in Q3b.

Summary

Configuration	2023 Backcast Revenue	Key Method
Standalone solar	\$18.910M	Hourly solar output × hourly RT price
Standalone storage	\$2.474M	Daily four-low / four-high arbitrage
Combined solar + storage	\$21.391M	Solar revenue plus storage with clipped solar charging first

The combined case produces only a small energy-arbitrage uplift over the standalone sum because the solar system clips very little energy in 2023. Figures above are energy-only; see Q2d for an illustrative reserve (headroom/footroom) scenario layered on top.